Logs traces and privacy

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Logs traces and privacy before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Failure budgets and service objectives

Without opening the old lesson, explain one key idea from Failure budgets and service objectives. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Integration tests and fixtures

Without opening the old lesson, explain one key idea from Integration tests and fixtures. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Least privilege and tool scopes

Without opening the old lesson, explain one key idea from Least privilege and tool scopes. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

> Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

Logs support diagnosis, but excessive recording can expose private content. **Trace ID**s link stages without storing complete sensitive **input**s.

For this policy log **trace ID**, stage, status, duration and **error category**; do not log passcodes, student text or teacher answers. Restrict access and declare retention.

Practise with fading help

> Complete the supported Logs traces and privacy trace and record the first step you can explain without help.

Use W2: Trace the worked case for Logs traces and privacy. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Logs traces and privacy explanation against the supplied facts before discussing it.

Use the worked case for Logs traces and privacy. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Logs contain user essay, teacher guide, stage=lookup, duration=12 ms and trace T8. Select allowed fields.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Logs traces and privacy facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

W2 Trace the worked case

> Trace the worked case. Apply the supplied Logs traces and privacy facts and keep your reasoning.

A failure record includes trace R7, status=denied and the complete student passcode. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

W3 Audit the limit

> Audit the limit. Apply the supplied Logs traces and privacy facts and keep your reasoning.

A report omits this qualification: A **trace ID** can still be linkable; **minimisation** and role-controlled access remain necessary. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Logs traces and privacy facts and keep your reasoning.

Logs contain user essay, teacher guide, stage=lookup, duration=12 ms and trace T8. Select allowed fields.

> **My reasoning and evidence**

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Logs traces and privacy facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Logs traces and privacy to Grounded Learning Service, identify a limit and choose the next useful test.

Add a labelled logs traces and privacy **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of logs traces and privacy.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

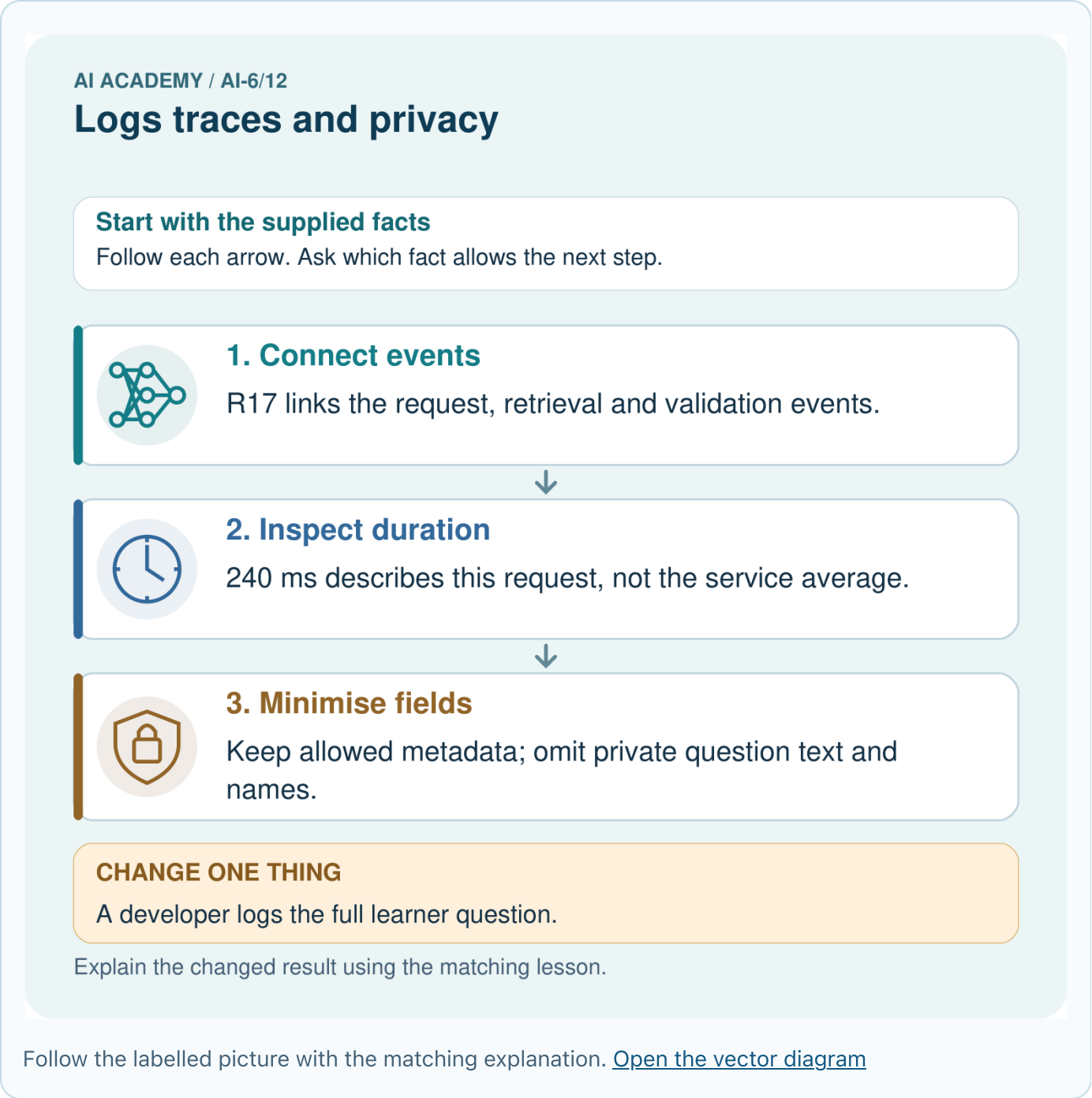
Pictorial case practice: Logs traces and privacy

> Trace without storing private text. A learner request is retrieved and checked under trace

> identifier R17. Explain the labelled step before checking the changed condition.

Trace without storing private text. A learner request is retrieved and checked under **trace id**entifier R17.

R17 is the request **trace ID**; C8 is the retrieved source ID. Allowed fields also include validation pass and duration 240 ms. Raw question text is excluded.



Follow the labelled picture with the matching explanation. Open the vector diagram

Identify the purpose of R17, C8 and 240 ms in the trace. Which field is deliberately omitted, and why would logging the whole question violate this design?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Project portfolio milestone

Keep this week's input, deciding step, result and limit together in the Grounded Learning Service record.

Project: Grounded Learning Service

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Trace without storing private text. A learner request is retrieved and checked under **trace id**entifier R17.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Keep allowed metadata; omit private question text and names.

1. Inspect duration

2. Connect events

3. Minimise fields

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Minimise fields Keep allowed metadata; omit private question text and names. Logs record events; traces connect related events.

Useful debugging fields should remain within the declared privacy policy.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: A developer logs the full learner question. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. That additional field violates this logging policy.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: That additional field violates this logging policy. Debugging convenience does not authorise storing more personal information.

Repair a missing building block

Review Failure budgets and service objectives (ai-6/11). Goal: Explain the **mechanism** and terminology of failure budgets and service objectives. Short practice: Calculate the maximum allowed failures among 100 requests.

Does 97 successes and 3 failures meet the target, and what changes after one failure becomes a success?. Return to this week and explain what you changed.

Review Integration tests and fixtures (ai-6/9). Goal: Explain the **mechanism** and terminology of integration tests and fixtures. Short practice: Follow request N through the fixture and adapter.

Which source and item should appear in the reply, and what precisely fails when the adapter returns M?. Return to this week and explain what you changed.

Review Least privilege and tool scopes (ai-6/6). Goal: Explain the **mechanism** and terminology of least privilege and tool scopes. Short practice: Classify read S2, write S2 and read S5 as permitted or denied.

Identify the operation or station mismatch for each denial, then reconsider after read S5 is added to trusted scope.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

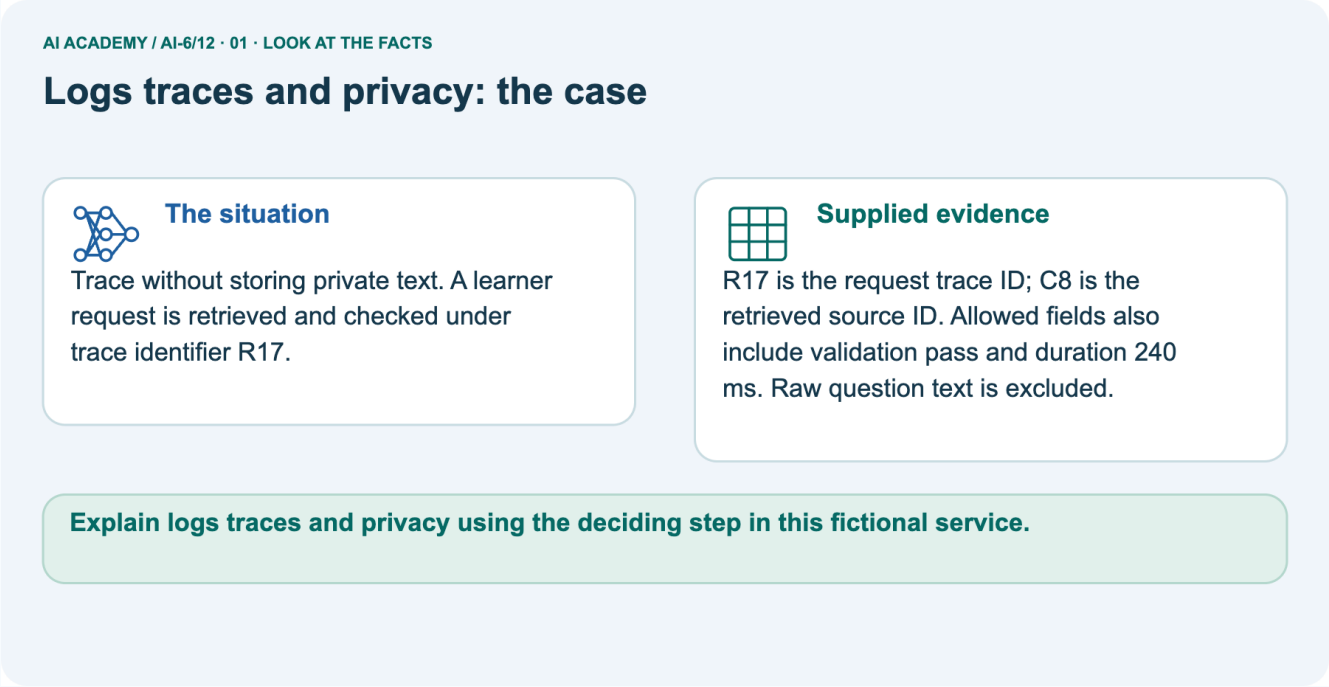
What I repaired, the evidence I kept and the next test

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Logs traces and privacy picture from its labels, then record a relevant limit.

01. Logs traces and privacy: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

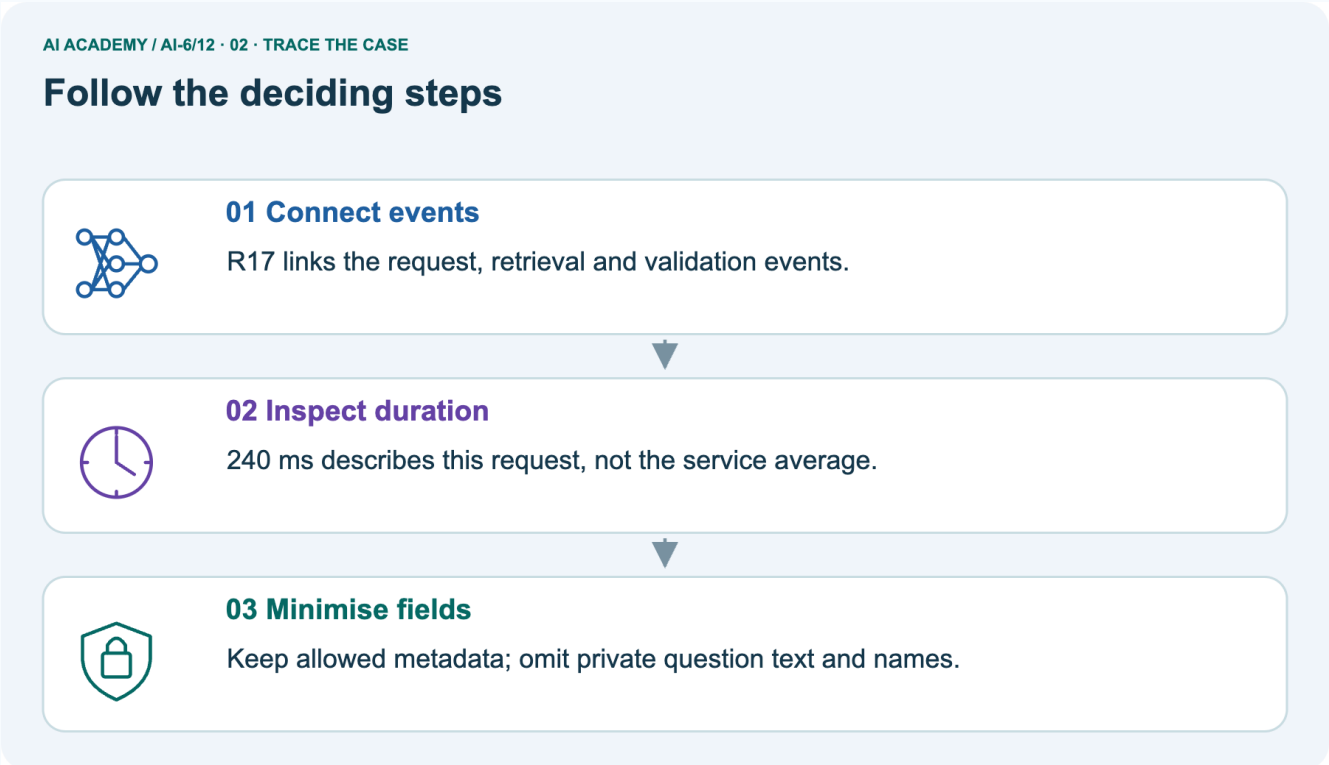
Trace without storing private text. A learner request is retrieved and checked under **trace id**entifier R17. Supplied facts: R17 is the request **trace ID**; C8 is the retrieved source ID.

Allowed fields also include validation pass and duration 240 ms. Raw question text is excluded.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



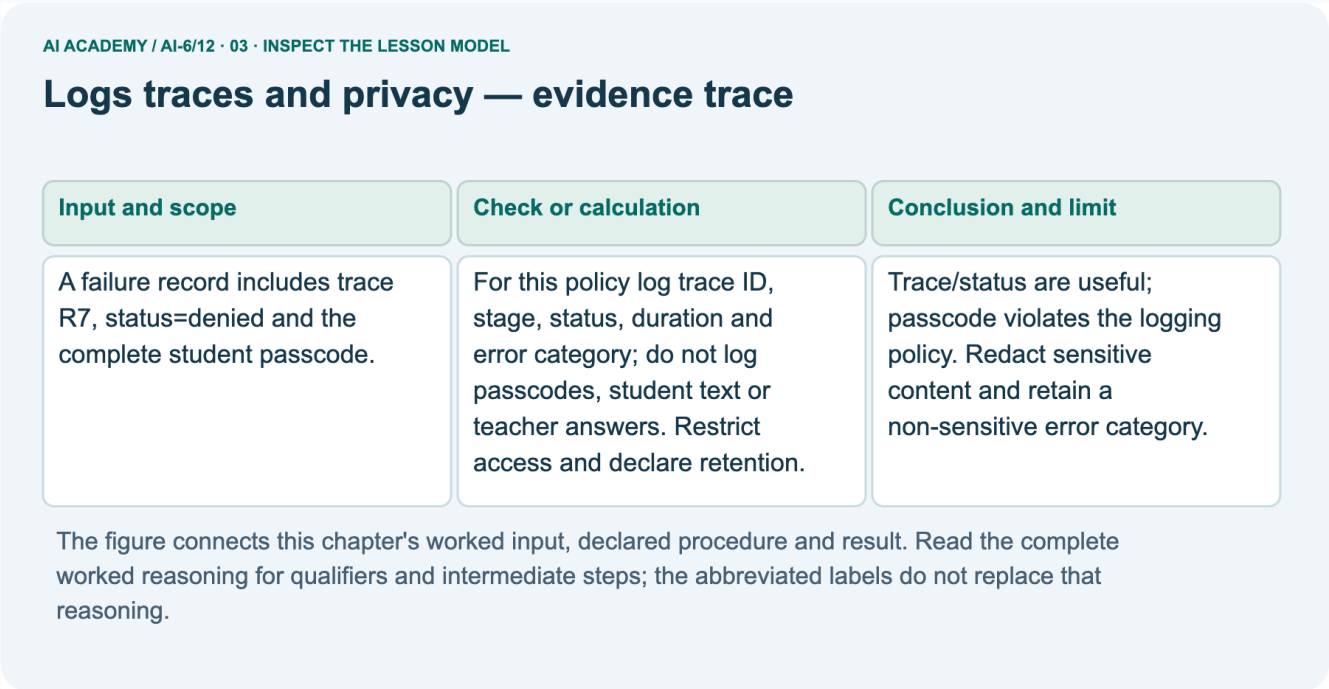
Follow the labelled picture with the matching explanation. Open the vector diagram

Connect events: R17 links the request, retrieval and validation events. → Inspect duration: 240 ms describes this request, not the service average. → Minimise fields: Keep allowed metadata; omit private question text and names.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Logs traces and privacy — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Logs traces and privacy — **evidence** trace.

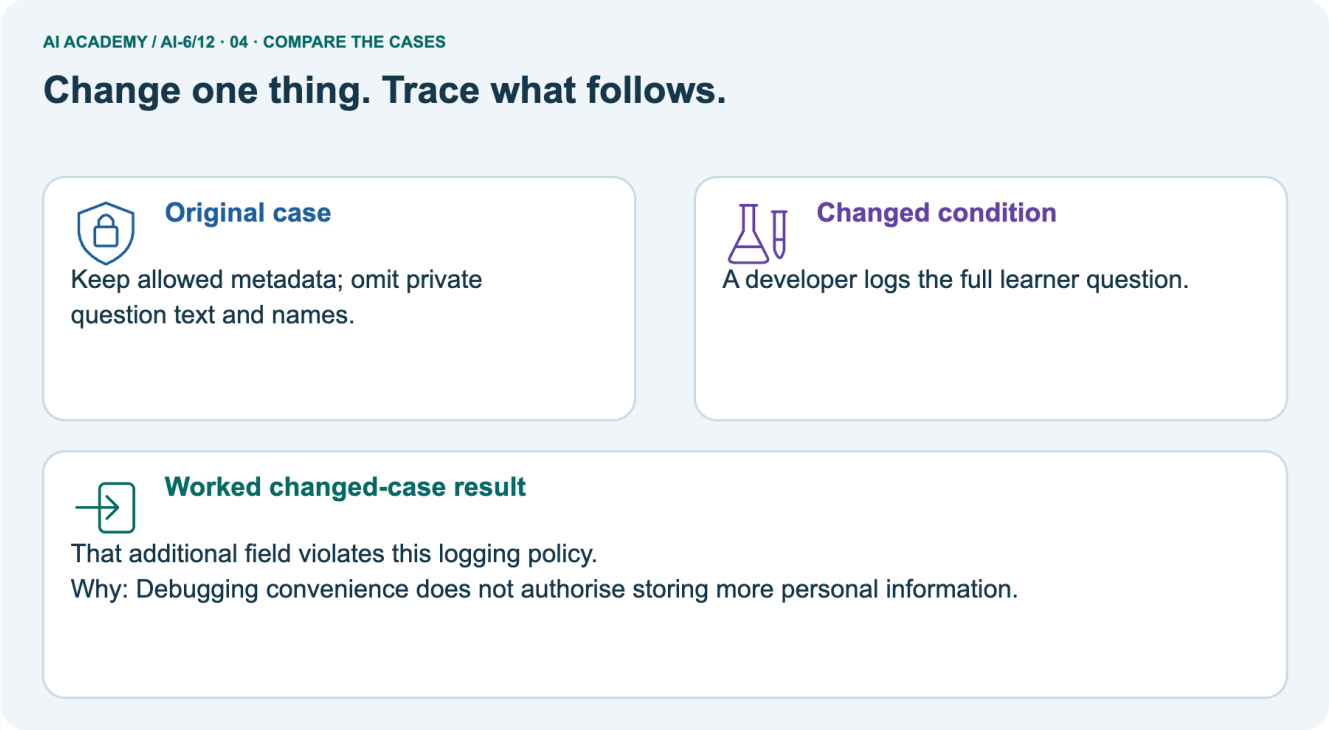
A failure record includes trace R7, status=denied and the complete student passcode. | For this policy log **trace ID**, stage, status, duration and **error category**; do not log passcodes, student text or teacher answers.

Restrict access and declare retention. | Trace/status are useful; passcode violates the logging policy. Redact sensitive content and retain a non-sensitive **error category**.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

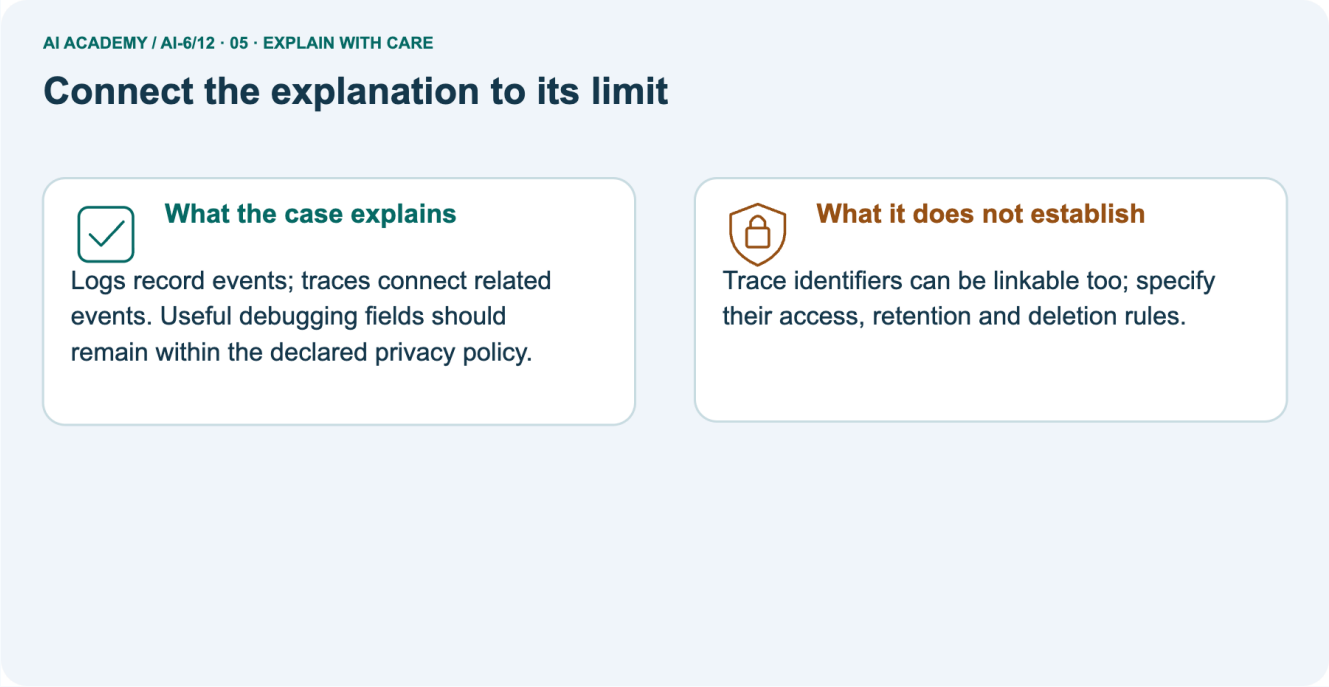
Original case: Keep allowed metadata; omit private question text and names. Changed condition: A developer logs the full learner question. Expected result: That additional field violates this logging policy.

Reason: Debugging convenience does not authorise storing more personal information.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



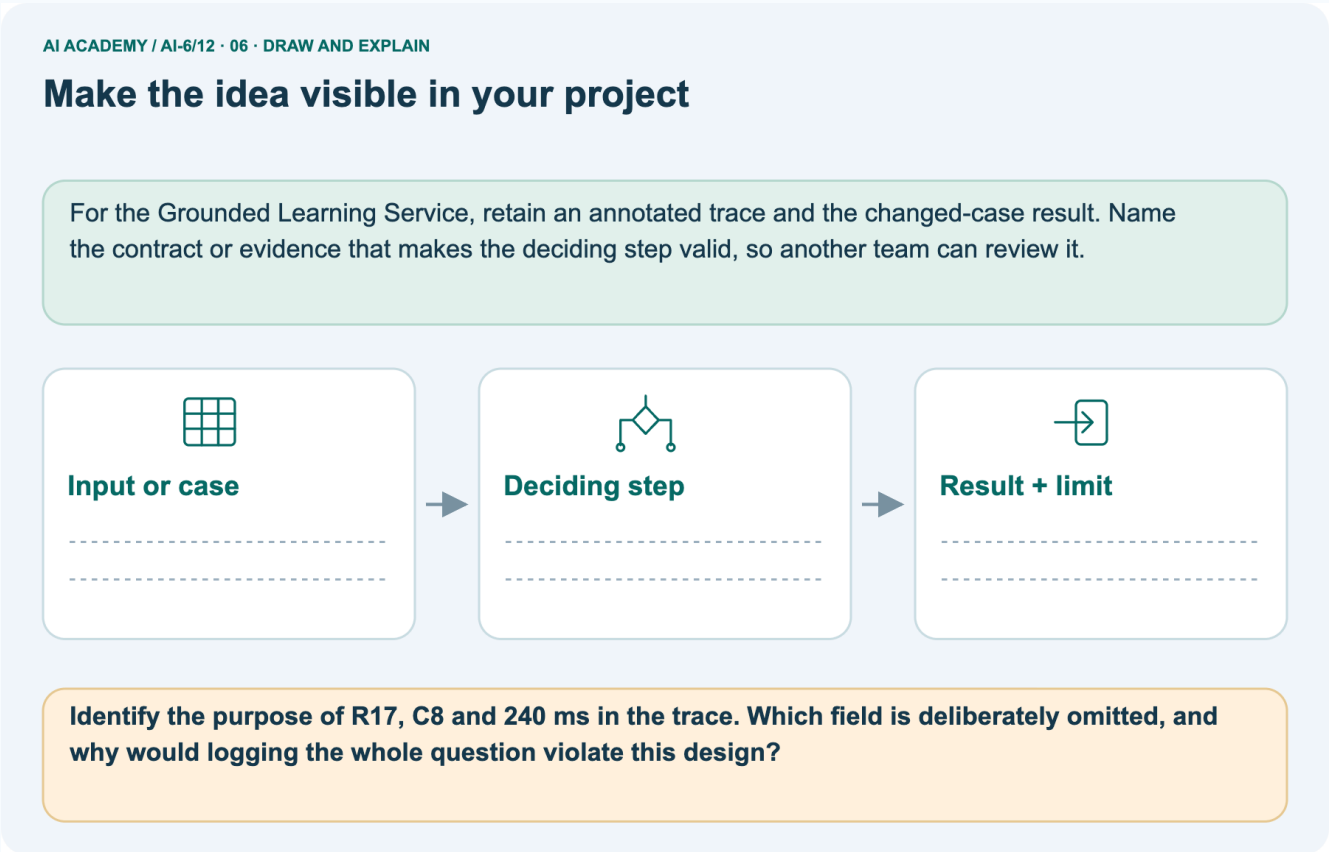
Follow the labelled picture with the matching explanation. Open the vector diagram

Logs record events; traces connect related events. Useful debugging fields should remain within the declared privacy policy. Limit: **Trace id**entifiers can be linkable too; specify their access, retention and deletion rules.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Identify the purpose of R17, C8 and 240 ms in the trace. Which field is deliberately omitted, and why would logging the whole question violate this design?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.